

ICT702 Data Visualisation

Purposeful Use of Colour

Module 4





I would like to acknowledge the traditional custodians of the
land we are meeting on,
The Gubbi Gubbi/Kabi Kabi peoples,
and pay my respect to Elders past, present and emerging.



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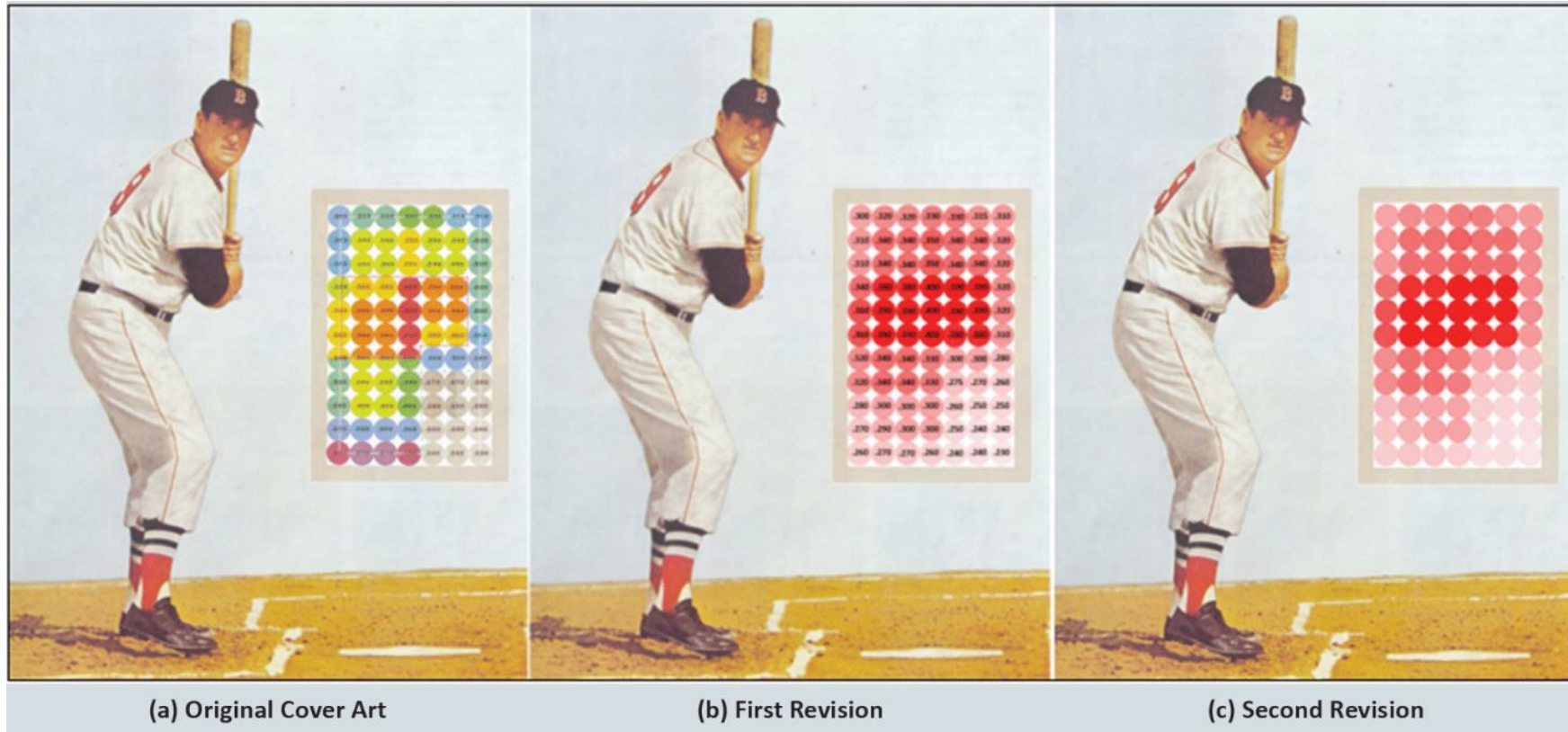
Learning Objectives for Module 4

- Describe hue, saturation, and luminance and differentiate between them.
- Describe the differences between colour psychology and colour symbolism
- Explain how each colour psychology and colour symbolism can be used effectively
- Design colour schemes that are appropriate for categorical data, ordered data, and quantitative data with meaningful reference values
- Define colours in data visualization software using the hue, saturation, luminance (HSL) system
- Create data visualizations using colours that are easier for the audience to interpret
- List common mistakes made when using colour in data visualizations and how to avoid them

Part 1

Data Visualization Makeover

Versions of cover art for Ted Williams's The Science of Hitting

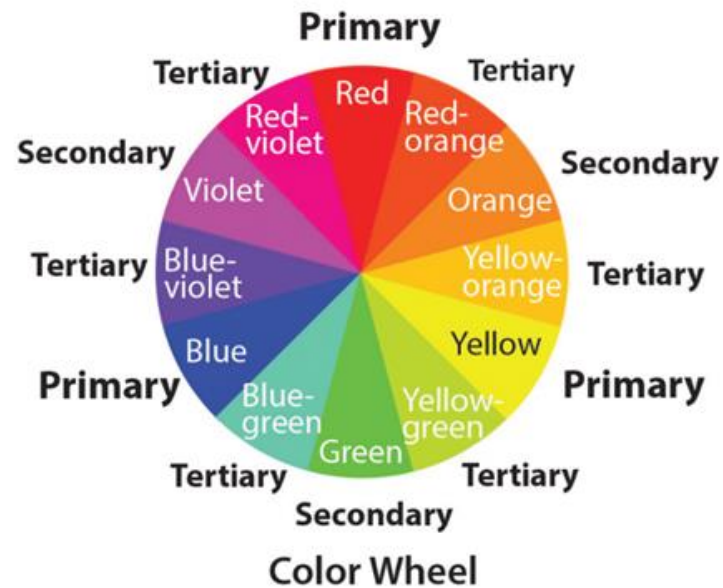
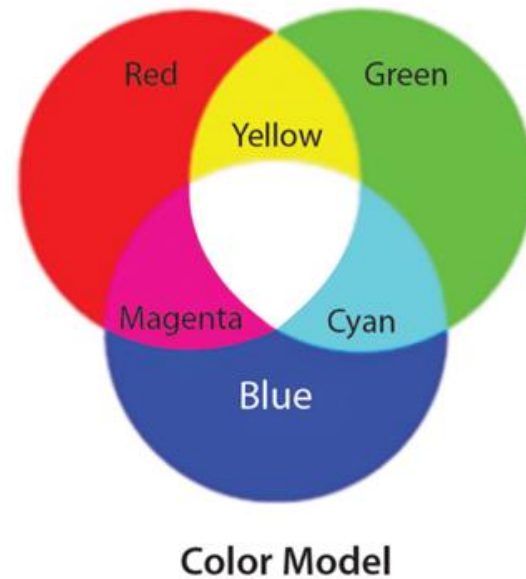


Colour

Colour is the property of an object that results from how the object reflects or emits light.

Attributes of Colour: Hue

The RGB primary colour model and colour wheel

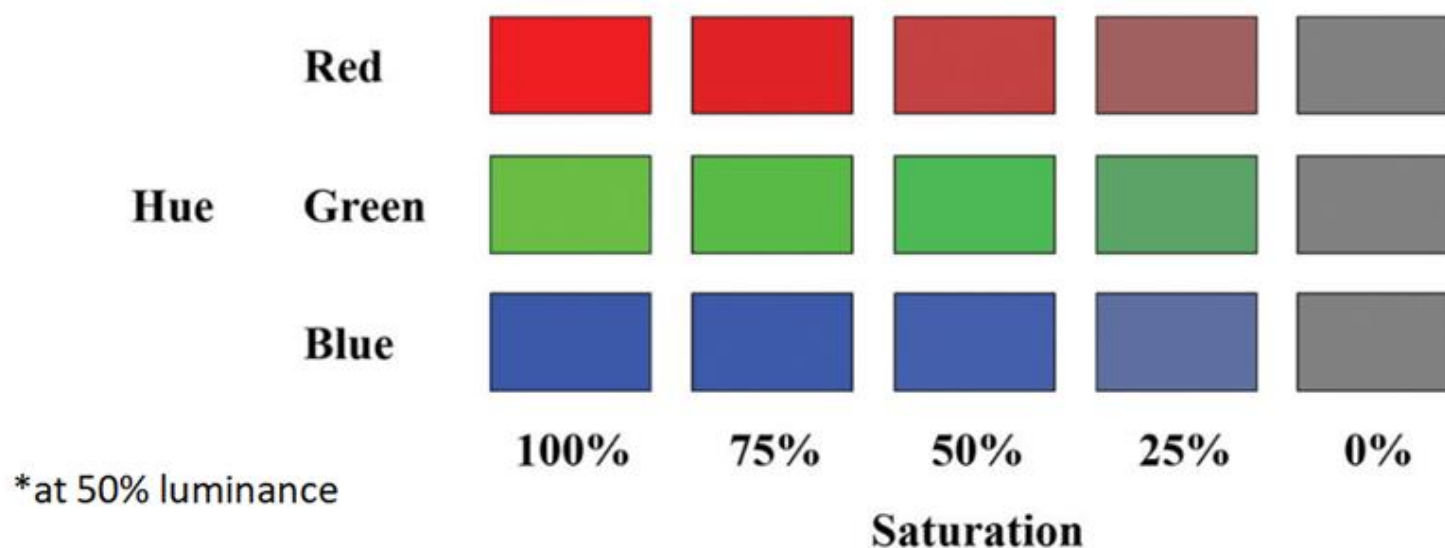


Hue is the base of colour. The **primary hues** (red, green, and blue) form the **colour model**. The primary hues cannot be mixed or formed by any combination of other hues.

Combinations of primary hues create secondary and tertiary colours, as displayed in the **colour wheel**.

Attributes of Colour: Saturation

Primary hues in the RGB primary colour model at different levels of saturation

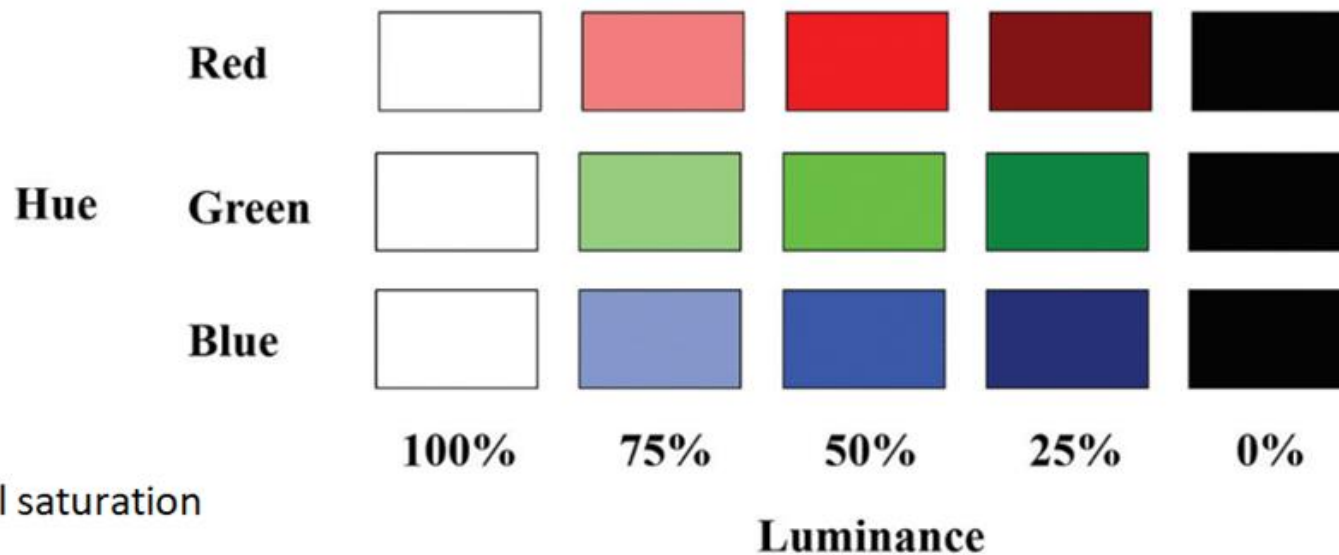


Saturation

- attribute of colour representing amount of gray present
- determines the hue's intensity or purity
- pure hue has no grayness and is 100% saturated

Attributes of Colour: Luminance

Primary hues in the RGB primary colour model at different levels of luminance



Luminance

- measures the relative degree of black or white within a colour – the brightness of a colour.
- adding white to a hue creates a brighter colour
- adding black to a hue creates a darker colour

Colour Psychology and Colour Symbolism

- **Colour psychology:** the study of the innate relationships between colour and human behavior.
 - We perceive **Cool hues** (**purple**, **blue**, and **green**) as soothing, calming, and reassuring.
 - **Warm hues** (**yellow**, **orange**, and **red**) evoke energy, passion, and danger.
- **Colour symbolism** refers to the cultural meanings and significance associated with colour.
 - Colour symbolism differs much more across cultures than colour psychology does and can change over time.

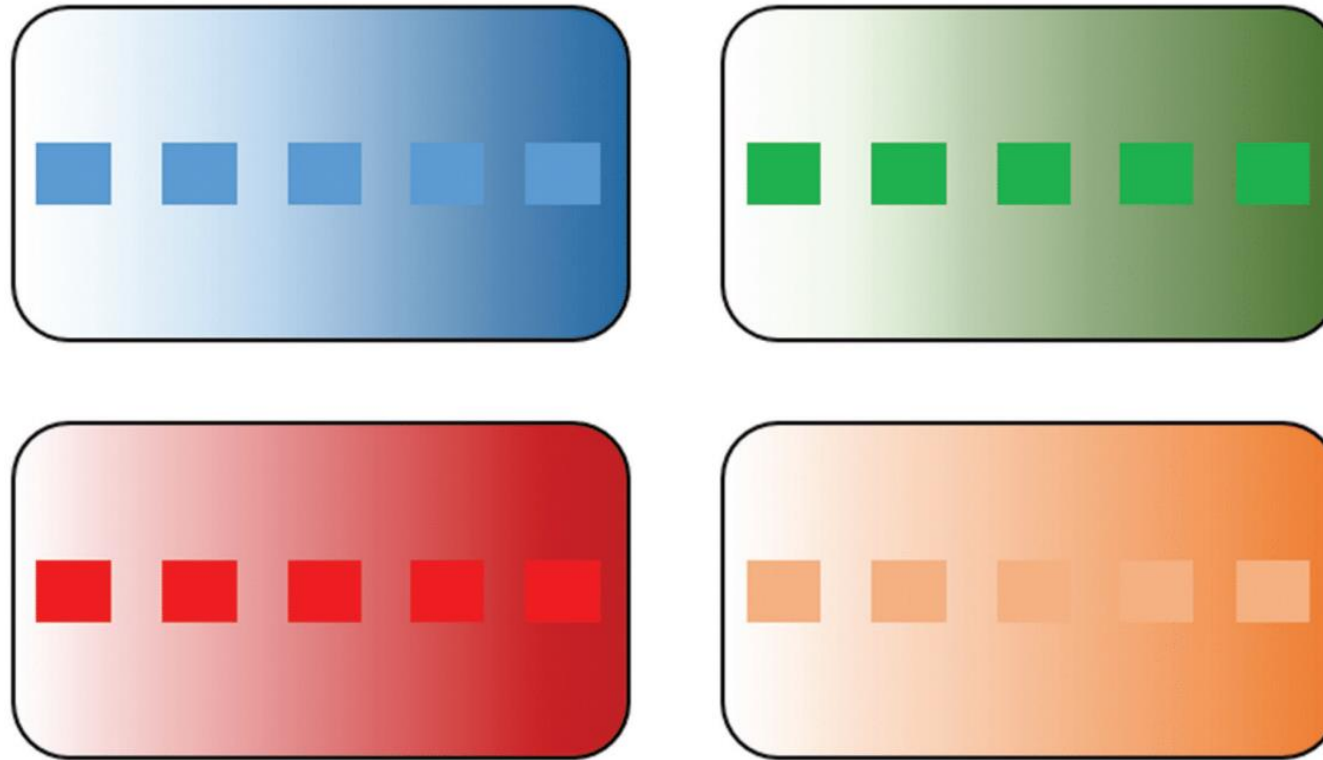
Perceived Colour

Traits frequently associated with various colours

COLOR	MEANINGS AND ASSOCIATIONS	COLOR	MEANINGS AND ASSOCIATIONS
Red	Anxiety, Arousing, Daring, Dominant, Energy, Excitement, Health, Life, Love, Passion, Power, Protection, Spirited, Stimulating, Strength, Up-To-Date	Purple	Authenticity, Charming, Dignified, Exclusive, Luxury, Quality, Regal, Sensuality, Sophistication, Spiritual, Stately, Upper Class
Orange	Abundance, Arousing, Comfort, Daring, Excitement, Extraversion, Fun, Happiness, Lively, Security, Sensuality, Spirited, Warmth	Pink	Charming, Cheerful, Feminine, Gentle, Nurturing, Sincerity, Soft, Sophistication, Tranquility, Warmth
Yellow	Arousing, Cheerful, Confidence, Creativity, Excitement, Extraversion, Friendliness, Happiness, Optimism, Self-Esteem, Sincerity, Smiley, Spirited	Brown	Nature, Outdoorsy, Reliability, Ruggedness, Security, Support, Tough
Green	Calm, Comfort, Equilibrium, Harmony, Health, Hope, Nature, Outdoorsy, Peace, Prosperity, Relaxation, Security, Serenity, Soothing, Tender	Black	Dignified, Efficiency, Elegance, Emotional Safety, Glamour, Power, Richness, Ruggedness, Security, Sophistication, Stately, Substance, Tough, Upper Class
Blue	Calm, Comfort, Competence, Coolness, Dignified Duty, Efficiency, Intelligence, Logic, Peace, Reflection, Relaxation, Reliability, Security, Serenity, Soothing, Successful, Tender, Tranquility, Trust	White	Calm, Clarity, Cleanness, Down-to-Earth, Happiness, Heavens, Honest, Hygiene, Innocence, Peace, Purity, Serenity, Sincerity, Soothing, Tender

Perceived Colour and Background

Perceived colour changes as contrast with the background colour changes



Complementary and Analogous Colours

Complementary colours are directly opposite each other on the colour wheel.

- Create dissonance
- Useful to make objects in a display stand out

Analogous colours are directly adjacent to each other on a colour wheel.

- Appear softer and smoother than complementary colours when used together

Overuse of complementary and analogous colours increases **cognitive load** and may distract the audience.

Suggestions for the Use of Colour in Data Visualization

- Poor and unnecessary use of colour creates clutter and increases the audience's cognitive load.
- A multicoloured background for data visualization can be distracting.
- Cool colours appear more distant than warm colours.
- Complementary colours used together create colour dissonance.
- Analogous colours create more colour harmony than complementary colours.

End of Part 1